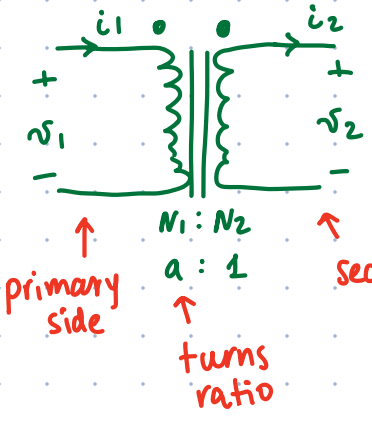
Review 3 (HW 3)

09/21/2026

Ideal transformers, Magnetic equiv. circuits

Ideal transformers:

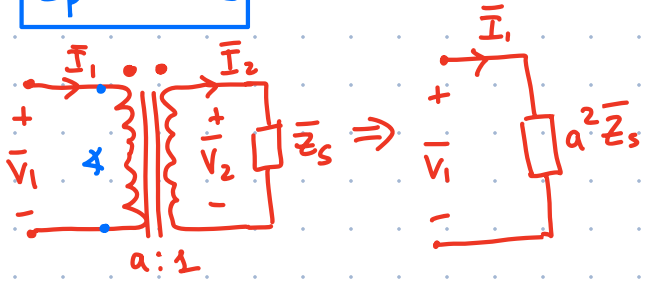


$$\frac{i_1}{i_2} = \frac{N_2}{N_1} = \frac{1}{a} \qquad \frac{v_1}{v_2} = \frac{N_1}{N_2} = a$$

$P_{in} = P_{out}$

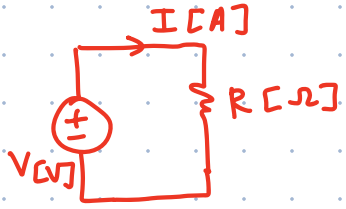
Moving impedance accross a transformer:

$$\bar{Z}_p = a^2 \cdot \bar{Z}_s$$

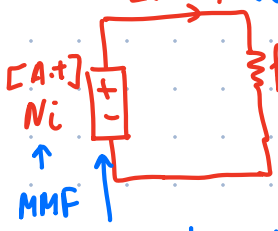


Mag. equiv. circuits:

Electrical



Magnetic



polarity: ϕ flows out of "+" into "−"

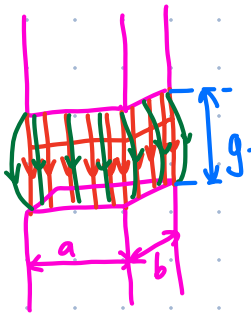
$$\mathcal{R} = \frac{l}{\mu_r \mu_0 A} \qquad (\mu \to \infty,\ \mathcal{R} \to 0)$$

if any of these changes, we need to introduce new reluctance

$\lambda = N\phi = Li$ (flux linkage) [Wb-t]

$v = \frac{d\lambda}{dt} = L\frac{di}{dt}$

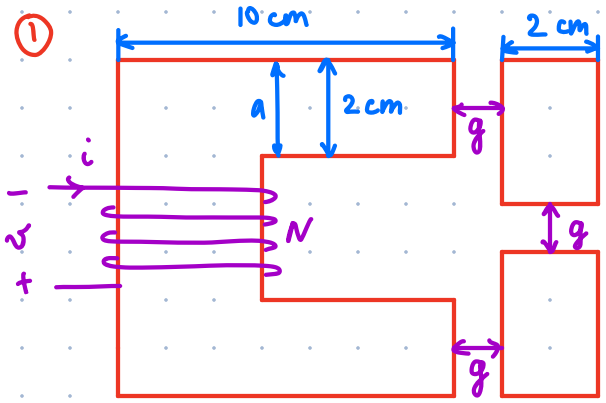
Fringing Effects:



No fringing: $A_g = a \cdot b = A_{core}$

Fringing: $A_g \uparrow$; $A_g = (a+g)(b+g)$

①



$g = 5\,mm$
$N = 250$
$\mu_{core} = \infty$
$\mu_0 = 4\pi \cdot 10^{-7}\ H/m$
$d\ (depth) = 3\,cm$

Fringing effects are included

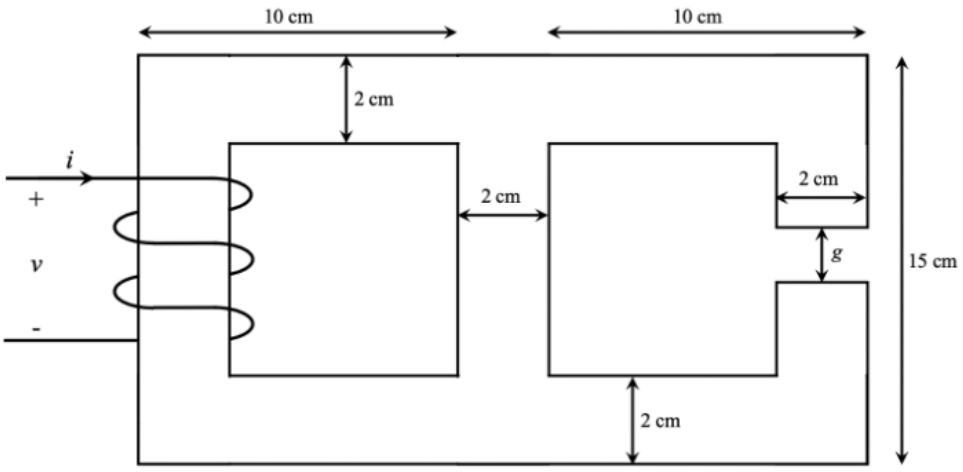
a) Area of the airgap (include fringing)?

b) Draw the magnetic equiv. circuit and find the reluctance of one airgap.

c) Find inductance of the structure, L.

d) Find inductance of the structure, Lnew, if fringing is neglected

② Fall 2019, M1, Pr 3



A coil is wrapped 300 times around an iron core with the given dimensions and a depth of 2 cm into the page. The iron has a permeability $\mu=2000\mu_0$ and contains an air gap of g=2 mm. Fringing effects can be neglected.

a) Draw the magnetic path through the iron core on the figure above and draw the corresponding magnetic equivalent circuit. (15 points).

b) What is the total flux ϕ through the circuit? (7 points)

c) What is the inductance of the coil? (3 points)